



EARTH AND SPACE SCIENCE 3

ACADEMIC

Grade 11/12

Curriculum Guide

EARTH AND SPACE SCIENCE 3
Prerequisites: Earth and Space Science 1, Earth and Space Science 2

Course Description:

Earth and Space Science 3 deepen the learners' understanding of the universe and its relevance to Earth systems through the study of cosmology, stellar systems, planetary motion, and space technologies. The course examines scientific evidence supporting models of the origin and evolution of the universe, the properties and life cycles of stars, and the detection of planets beyond the solar system. The learners investigate how astronomical observations and space-based technologies contribute to scientific discovery and environmental monitoring.

Building on these concepts, the course explores how satellite observations and climate data are used to analyze climate change trends and impacts in the Philippines. The learners examine the causes of climate change, evaluate national and international climate policies, and assess mitigation and adaptation strategies that strengthen community resilience. Through data analysis, modeling, and project-based investigations, the learners apply scientific reasoning and digital tools to interpret Earth and space data and propose evidence-based solutions for environmental sustainability and climate action.

Elective: Academic

Time Allotment: 80 hours for one term

CONTENT	CONTENT STANDARDS <i>The learners demonstrate understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
Origin of the Universe, Stellar Systems, and Applications of Space Technology		
1. Origin and evolution of the universe	1. observational evidence such as galaxy recession rates, cosmic microwave background radiation, and the large-scale distribution of matter provides scientific support	1. evaluate competing cosmological models using observational evidence, including galaxy recession rates, the age of the universe, and the large-scale structure of matter;
		2. use redshift data and cosmic microwave background evidence to evaluate the Big Bang model;

CONTENT	CONTENT STANDARDS <i>The learners demonstrate understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
	for models explaining the origin and evolution of the universe;	3. analyze the timeline from the Big Bang to the formation of the first stars and galaxies;
2. Stellar properties and planetary motion	2. stars have measurable properties and life cycles that can be studied through spectral analysis and classification, and that planetary motions follow predictable patterns governed by gravitational laws;	4. analyze stellar spectra to classify stars using the Hertzsprung-Russel diagram;
		5. apply Kepler's laws to explain planetary orbital characteristics and habitability zones;
3. Exoplanets and astrobiology	3. various observational techniques enable the detection of planets beyond the solar system and provide evidence that helps scientists evaluate the possibility of life in other planetary systems;	6. conduct investigation on the exoplanet detection methods and their relevance to astrobiology;
		7. evaluate evidence from exoplanet research (e.g., atmospheric composition, temperature, and distance from a star) to assess conditions for extraterrestrial life;
4. Space technology and Earth observation	4. advances in space exploration technologies enhance our understanding of Earth and other planets; and 5. satellite technology and remote sensing enable scientists and government agencies to observe Earth systems, monitor environmental changes, and support disaster risk reduction and environmental management.	8. identify key space technologies for space exploration and Earth observation and their primary functions;
		9. use satellite imagery (e.g. from PhilSA Space Data Dashboard, NASA Worldview or Copernicus Browser) to analyze land cover change, deforestation, or urban expansion in a Philippine locality over time; and
		10. evaluate how PhilSA and international space agencies use remote sensing data to monitor Philippine environment conditions and support disaster response.

CONTENT	CONTENT STANDARDS <i>The learners demonstrate understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
Performance Standards <i>The learners can analyze observational and satellite data to explain models of the universe, classify stars and planetary systems, and evaluate evidence related to the existence of exoplanets and extraterrestrial life. They can apply scientific reasoning and digital tools to interpret astronomical and Earth observation data and communicate how space science and remote sensing technologies contribute to environmental monitoring and disaster risk management in the Philippines.</i>		
Suggested Performance Tasks • Create a scientific report, infographic, or digital presentation explaining how exoplanet detection methods work, providing examples of discovered exoplanets, describing characteristics that may support habitability, and discussing implications for the search of extraterrestrial life. • Produce a map-based report or multimedia presentation that compares satellite images from different years, identifies changes in land cover (e.g., deforestation, urban expansion, coastal changes), explains possible causes of the changes, and discusses how remote sensing supports environmental monitoring and disaster preparedness.		
Climate Change and Climate Action		
5. Space technologies for climate observation	6. satellite observations and remote sensing technologies provide critical data used to monitor Earth's climate system and environmental changes;	11. use satellite-derived datasets including temperature anomalies, sea level measurements, and atmospheric CO ₂ to identify climate trends in the Philippines;
		12. use satellite and ocean observation data to analyze the sea level rise and its potential impacts on Philippine coastal communities;
6. Causes of climate change	7. climate change results from natural processes and human activities that alter the balance of energy in Earth's climate system;	13. explain how enhanced greenhouse effect and human activities contribute to global warming and climate change;
		14. calculate personal or school-level carbon footprints and relate the results to national greenhouse gas emissions data;

CONTENT	CONTENT STANDARDS <i>The learners demonstrate understanding that...</i>	LEARNING COMPETENCIES <i>The learners...</i>
7. Climate policy and governance	8. international agreements and national policies guide efforts to mitigate climate change and support adaptation strategies; and	15. describe the commitments of the Philippines under the Paris Agreement and their role in reducing greenhouse gas emissions;
		16. analyze the provisions of Republic Act No. 9729 and explain how local government units implement climate adaptation plans;
8. Climate resilience and sustainable action	9. climate mitigation and adaptation strategies can strengthen community resilience and reduce the impacts of climate change.	17. describe the mitigation and adaptation strategies used in Philippine communities; and
		18. create a community-level climate resilience plan integrating renewable energy planning, waste reduction, and ecosystem-based adaptation.
Performance Standard <i>The learners can analyze satellite and climate datasets to explain climate change trends and their impacts on the Philippines. They can apply scientific reasoning and data analysis to evaluate climate mitigation and adaptation strategies and communicate evidence-based recommendations for strengthening community climate resilience and sustainability.</i>		
Suggested Performance Tasks • Prepare a scientific report or digital presentation that identifies climate trends in the Philippines, explains possible causes of the observed trends, and discusses potential environmental and societal impacts. • Design and present a climate resilience plan for the school or a local community that evaluates climate risks, proposes mitigation and adaptation strategies, integrates renewable energy and ecosystem-based solutions, and demonstrates how national policies (Republic Act No. 9729) and international agreements (Paris Agreement) guide climate action.		